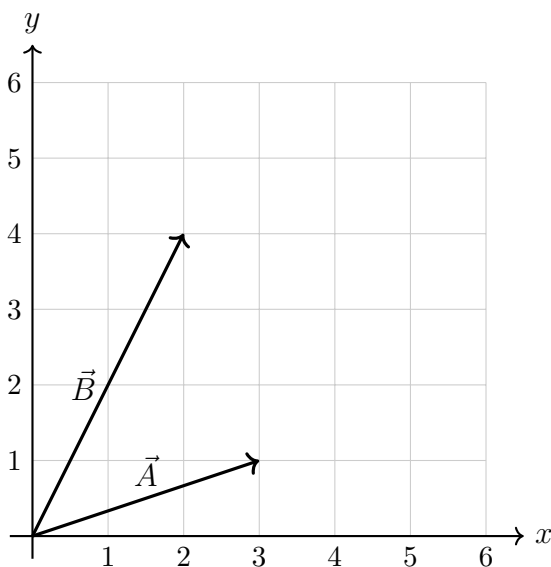


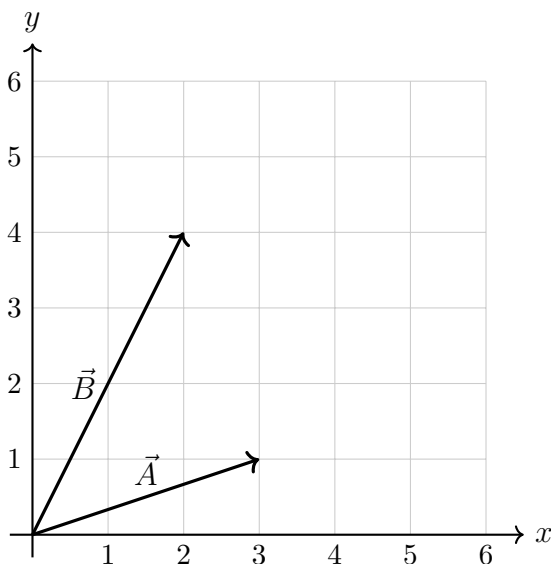
3.1 Classwork: Vector Addition

1. Two vectors $\vec{A}$ and $\vec{B}$ are drawn from the origin. Using the **head-to-tail method**, find the resultant $\vec{R} = \vec{A} + \vec{B}$. Show your construction and draw the resultant on the grid.



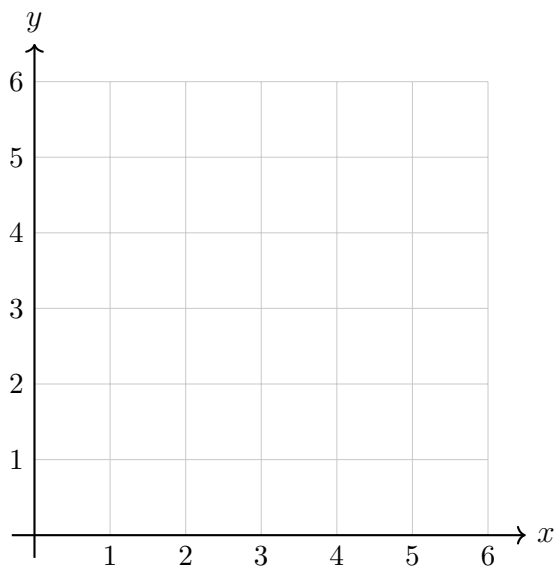
$$\vec{R} = (\quad , \quad)$$

2. The same two vectors are drawn below. Using the **parallelogram method**, find the resultant $\vec{R} = \vec{A} + \vec{B}$. Complete the parallelogram and draw the resultant as the diagonal from the origin.



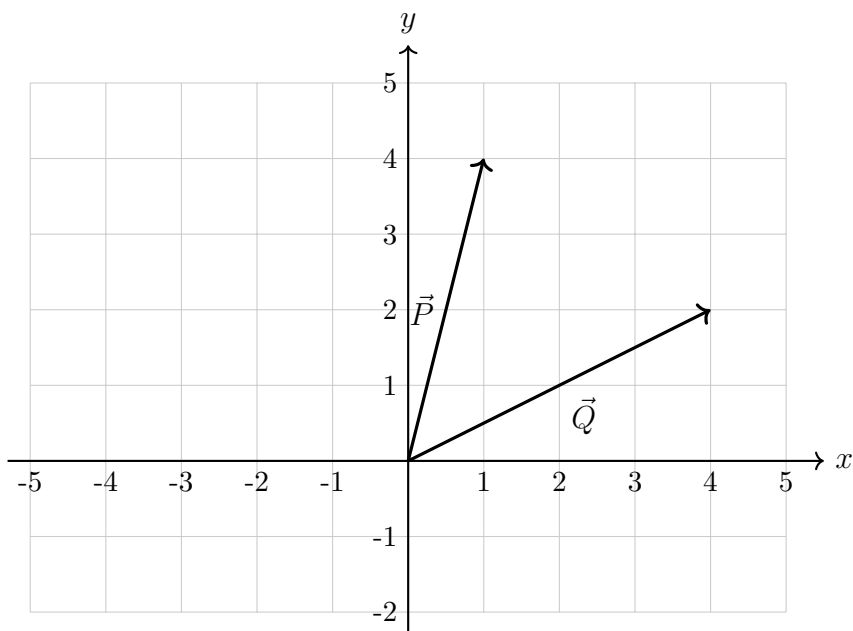
$$\vec{R} = (\quad , \quad)$$

3. Draw vectors $\vec{C} = (4, 2)$ and $\vec{D} = (1, 3)$ from the origin on the grid below. Find $\vec{R} = \vec{C} + \vec{D}$ using either the head-to-tail or parallelogram method. Show your construction and draw the resultant.



$$\vec{R} = (\quad , \quad)$$

4. Two vectors $\vec{P}$ and $\vec{Q}$ are drawn on the grid below. Find $\vec{P} - \vec{Q}$ geometrically: draw the opposite of $\vec{Q}$ (that is, $-\vec{Q}$), then add it to $\vec{P}$ using the head-to-tail method. Draw the resultant.



$$\vec{P} - \vec{Q} = (\quad , \quad)$$